

B.TECH
(SEM V) THEORY EXAMINATION 2022-23
CHEMICAL REACTION ENGINEERING -II

Time: 3 Hours**Total Marks: 100****Note:** Attempt all Sections. If require any missing data; then choose suitably.**SECTION A****1. Attempt all questions in brief.****2 x 10 = 20**

- (a) Define void fraction of a catalyst
- (b) Catalyst activity and selectivity
- (c) Define catalyst inhibitors.
- (d) Define Solid density and particle density
- (e) Define Effectiveness factor
- (f) Write the steps of a catalytic reaction
- (g) Enhancement factor
- (h) Define Hatta modulus
- (i) Define molar density.
- (j) Define Biochemical reactions

SECTION B**2. Attempt any three of the following:****10 x 3 = 30**

- (a) Differentiate between physical adsorption and chemical adsorption.
- (b) Given Mass of catalyst = 101.5 gm, vol. of Helium displaced = 45.1 cm³, vol. of Hg displaced = 82.7 cm³. Calculate the pore volume, porosity, solid density and particle density?
- (c) Explain shrinking core model with the help of diagram.
- (d) Explain Thiele Modulus and its importance.
- (e) Discuss the heat effects during catalytic reaction

SECTION C**3. Attempt any one part of the following:****10 x 1 = 10**

- (a) What are different catalyst preparation methods? Explain any one method.
- (b) Explain catalyst poisoning and discuss its various types.

4. Attempt any one part of the following:**10 x 1 = 10**

- (a) Derive relation between t/τ and X_B for SCM for chemical reaction control.
- (b) Solid particles of B are roasted with gas A to give a firm non-flaking product according to SCM, $A(g) + B(s) \rightarrow R(g) + S(s)$, $C_A = 0.01$ kmol/m³, molar density = 20 kmol/m³. From the following data, find the rate controlling mechanism.

dp, mm	X_B	T, min
1	1	4
1.5	1	6

5. Attempt any one part of the following:

10 x 1 = 10

- (a) Derive the overall rate expression for Cumene decomposition into Benzene and Propylene, assuming single site mechanism and adsorption step be rate controlling.
- (b) For the reaction $A \rightarrow 4R$, $-r_A = 96 C_A$ mol/kg cat.hr, mixed flow reactor is used for 35% conversion of A to R for a feed rate of 2000 mol/hr of 80% pure A at 3.2 atm and 117° C. If $C_{A0} = 0.1$ mol/l, determine the amount of catalyst needed.

6. Attempt any one part of the following:

10 x 1 = 10

- (a) Discuss various kinetic regimes of mass transfer and reaction for fluid-fluid reactions.
- (b) Spherical particles of zinc blende of size $R = 1$ mm are roasted in an 8% oxygen stream at 900° C and 1 atm. The reaction is $2ZnS + 3O_2 \rightarrow 2ZnO + 2SO_2$. Assuming that reaction proceeds by the shrinking core model calculate the time needed for complete conversion of a particle and the relative resistance of ash layer diffusion during the operation. Note that film resistance can safely neglected as long as a growing ash layer is present.
 $\rho_B = 4.13$ gm/cm³, $k'' = 2$ cm/sec, diffusion coefficient = 0.08 cm²/sec.

7. Attempt any one part of the following:

10 x 1 = 10

- (a) A feed consisting 30% 50µm, 40% 100 µm and 30% 200 µm radius particles, is to be fed continuously in a PFR. The time required for complete conversion is 5, 10 and 20 min for the three sizes of particles. Find the conversion of solids for a residence time of 8 min in the reactor.
- (b) Derive Michaelis Menten equation and explain under which conditions it is valid. What is its importance?